

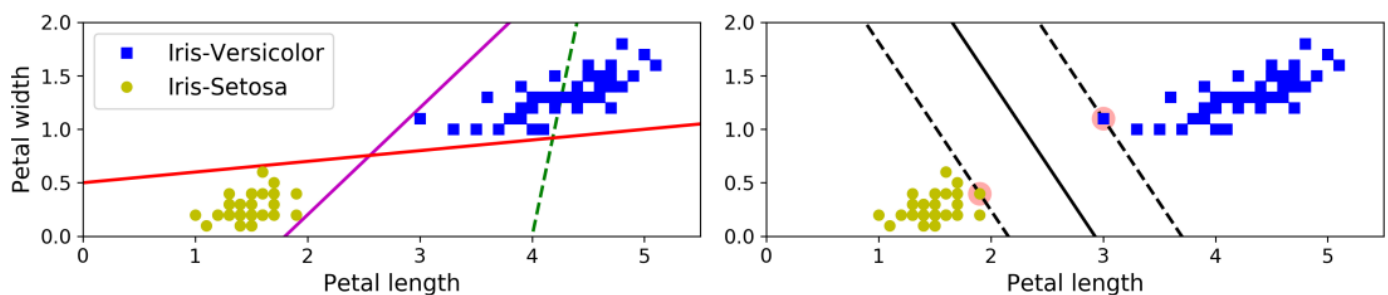
Support Vector Machines

Chapter 5: Support Vector Machines
Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow, Gurelien Geron, 3e, O'Reilly Media, 2022.

- A Support Vector Machine (SVM) is a very powerful and versatile Machine Learning model, capable of performing classification, regression, and outlier detection.
- SVMs are particularly well suited for classification of complex but small- or medium-sized datasets. However, they do not scale very well to very large datasets.
- Support Vector Machines classify data by finding a hyperplane that maximizes the margin between different classes in the training data.
- The objective of an SVM is to find the hyperplane that best separates the classes. Therefore, SVMs are also called large margin classifiers.
- Unlike logistic regression, a standard SVM classifier does not directly output probabilities. Instead, it predicts the class label of the input sample.
- An SVM can be linear or nonlinear.
- In a linear SVM, the data is linearly separable and can be divided into classes using a straight line (in 2D) or a hyperplane (in higher dimensions).
- In a nonlinear SVM, the data cannot be separated using a straight line or hyperplane. In such cases, kernel functions are used to transform the data into a higher-dimensional space where separation becomes possible.
- SVMs are sensitive to feature scales; therefore, feature scaling is usually performed before training the model.

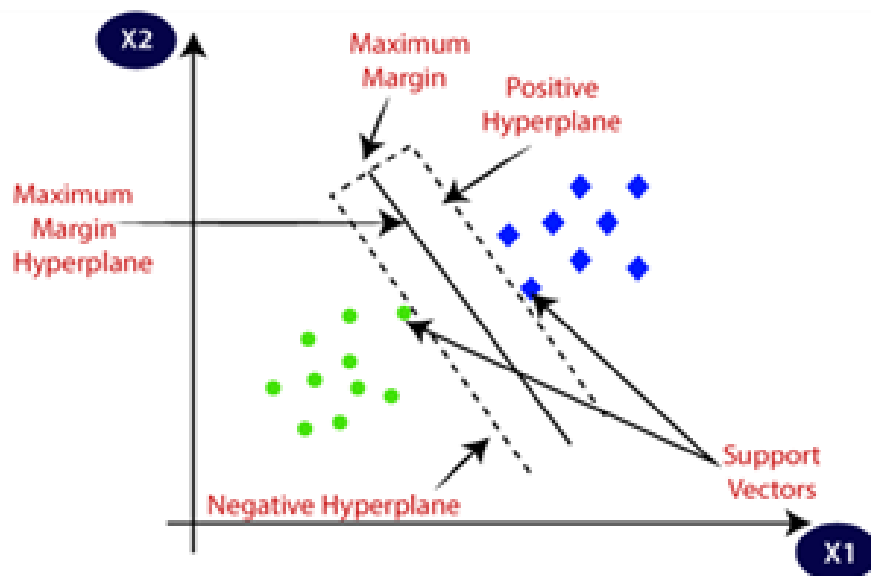
Example

- Consider the Iris dataset from the Scikit-learn toy datasets, containing two linearly separable classes of iris flowers: Iris-Setosa and Iris-Versicolor.
- The left plot shows the decision boundaries of three possible linear classifiers. Different learning algorithms (such as logistic regression, different linear neural network architectures) can produce different classifiers. The dashed-line classifier performs poorly because it does not separate the classes correctly. The other two classifiers perfectly separate the training data, but their decision boundaries lie very close to the training instances. Such classifiers have small margins and may not generalize well to unseen data.
- In contrast, the solid line in the plot on the right represents the decision boundary of an SVM classifier. This boundary not only separates the two classes correctly, but also maximizes the margin by staying as far away as possible from the nearest training instances (support vectors).



Linear SVM Classification

- A Linear Support Vector Machine (Linear SVM) is a supervised learning algorithm used for binary classification of linearly separable or nearly linearly separable data.
- The goal of a linear SVM is to find the optimal hyperplane that separates the classes with the maximum possible margin.
- The hyperplane is the decision boundary that separates the data points belonging to different classes.
- In a two-dimensional feature space, the hyperplane is a straight line. In higher dimensions, it becomes a plane or hyperplane.
- Linear SVM attempts to maximize the distance between the hyperplane and the nearest data points from both classes. This distance is called the margin.
- The training samples closest to the decision boundary are called support vectors. These points are critical because they determine the position of the optimal hyperplane.
- A larger margin generally improves the generalization ability of the classifier. Therefore, SVM searches for the hyperplane having the maximum margin, called the Maximum Margin Hyperplane (MMH).
- In a two-dimensional example with two classes, the hyperplane can be visualized as the widest possible straight “band” separating the two classes.



Hyperplane

- Formally, a hyperplane is an $m - 1$ subspace in an m -dimensional space.
- For example:
 - In a two-dimensional space, the hyperplane is a one-dimensional line.
 - In a three-dimensional space, the hyperplane is a two-dimensional plane.
- A hyperplane is therefore a generalized decision boundary in higher-dimensional spaces.

Support Vectors

- Support vectors are the training samples closest to the hyperplane.
- These points directly determine the position and orientation of the decision boundary.
- The margin is formed using these nearest points; therefore, only the support vectors influence the optimal hyperplane.
- Unlike logistic regression, where all training samples contribute to the decision boundary, SVM mainly depends on the support vectors. Adding more training instances outside the margins will not affect the decision boundary at all.

Margin

- The margin is the perpendicular distance between the hyperplane and the nearest training samples (support vectors).
- In SVM, a large margin is considered desirable because it generally improves classification performance and generalization.
- The objective of SVM training is to identify the support vectors and construct the hyperplane that maximizes this margin.
- There are two types of margins in SVM: hard margin and soft margin.

Hard Margin Classifier

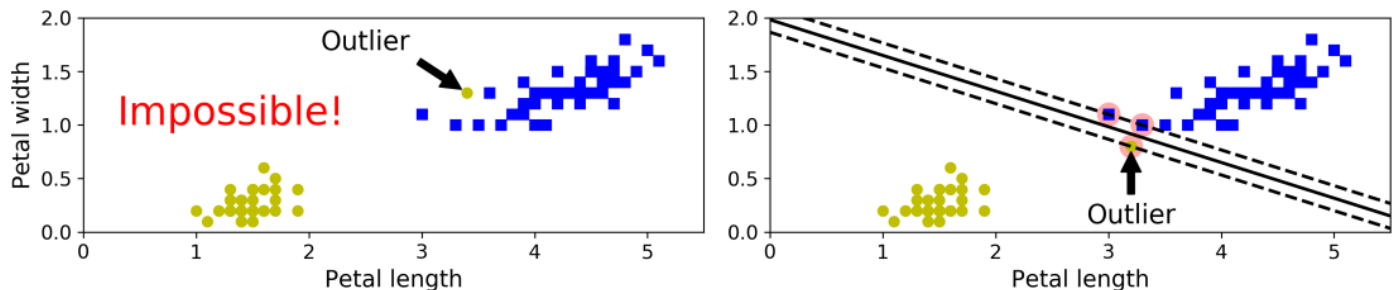
- A hard-margin SVM assumes that the training data is perfectly linearly separable.
- It tries to find a hyperplane that separates the classes with the maximum possible margin and zero classification error.
- No training samples are allowed inside the margin boundaries.
- Hard-margin SVM works well only when the data is perfectly separable and contains no noise or outliers.
- Also, it is quite sensitive to outliers.

Example

The following figure shows the iris dataset with just one additional outlier.

On the left, it is impossible to find a hard margin.

On the right the decision boundary ends up very different from the one we saw previously without the outlier, and it will probably not generalize as well.



- To avoid these issues it is preferable to use a more flexible model with soft margin classification, where instances may end up in the middle of the margin or even on the wrong side.

Soft Margin Classifier

- A soft-margin SVM allows some training samples to violate the margin constraints.
- It introduces flexibility so that a better generalization boundary can be found even when the data is noisy or overlapping.
- Some points may lie inside the margin, or even be misclassified.
- The objective is to find a good balance between keeping the margin as large as possible and limiting the margin violations.

Hypothesis Function / Prediction Function in a Linear SVM

- For a dataset having m input features and n samples, the equation of the hyperplane is:
$$w^T x + b = 0$$

Where,

$w = (w_1, w_2, \dots, w_m)$ is the weight vector,

$x = (x_1, x_2, \dots, x_m)$ is the input feature vector,

b = bias term.

- The hyperplane divides the feature space into two regions corresponding to the two classes.
- Any sample point that lies above the hyperplane satisfies $w^T x + b \geq 0$, and is classified as class +1.
- Similarly, any sample point that lies below the hyperplane satisfies $w^T x + b < 0$, is classified as class -1.
- In the SVM literature, it is common to assume the class labels are -1 and $+1$, rather than 0 and 1 .
- Hence the hypothesis / prediction function for linear SVM is:

$$h(x) = \hat{y} = \text{sign}(w^T x + b)$$

- The margin is the perpendicular distance between the separating hyperplane and the nearest training samples.
- The support vectors lie on the margin boundaries.
- The margin boundaries are given by:

$$w^T x + b = 1 \text{ and } w^T x + b = -1$$

These equations represent the margin boundaries.

Objective Function of Linear SVM

- There are two objectives for a linear SVM:
 - Maximize the margin
 - Penalize incorrect classifications and points lying inside the margin
- The width of the margin is inversely proportional to the weight vector w .
- The bias term b has no influence on the size of the margin: tweaking it just shifts the margin around, without affecting its size.
- Hence to maximize margin (width), we need to keep w as small as possible. We use:

$$\text{minimize } \frac{1}{2} \|w\|^2$$

Where $\|w\|$ is L2 norm and is equivalent to $\sqrt{w_1^2 + w_2^2 + \dots + w_m^2}$.

Squaring removes the square root, making the optimization mathematically easier to solve.

The factor of $1/2$ is added for convenience and doesn't affect the optimal parameters.

- Linear SVM uses the hinge loss function to penalize incorrect classifications and points lying inside the margin.
- It encourages training samples to lie on the correct side of the decision boundary with a margin of at least 1.
- The hinge loss for i^{th} training example is:

$$L(y^{(i)}, \hat{y}^{(i)}) = \max(0, 1 - y^{(i)}(w^T x^{(i)} + b))$$

- If a sample is correctly classified and lies outside the margin, the loss becomes zero.
 - If a sample lies inside the margin or is misclassified, a positive loss is incurred.
- Hence the objective function of linear SVM is:

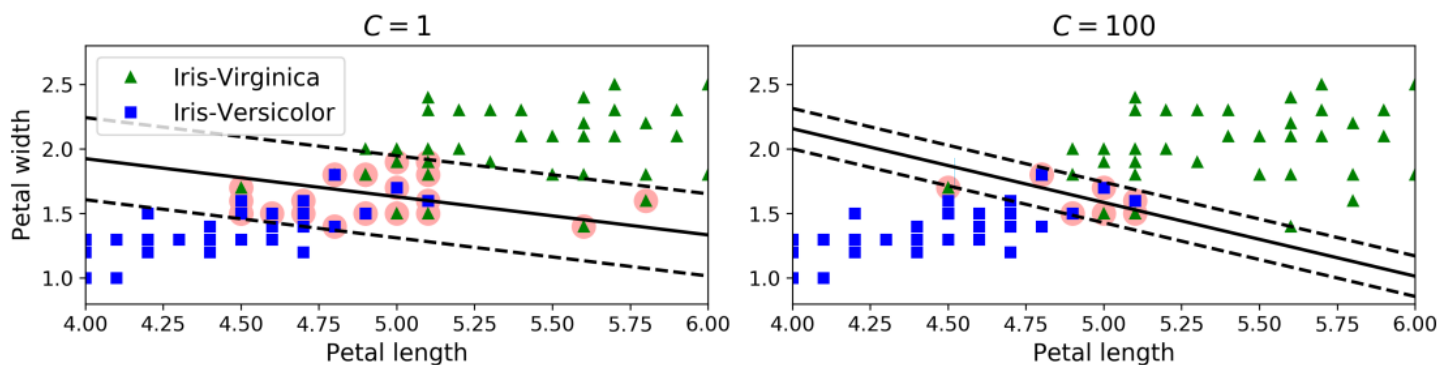
$$J(w, b) = \text{minimize } \frac{1}{2} \|w\|^2 + C \sum_{i=1}^n \max(0, 1 - y^{(i)}(w^T x^{(i)} + b))$$

Where $C \geq 0$ is the regularization parameter controlling the trade-off between margin maximization and classification error.

- This objective function is a convex quadratic optimization problem.

Regularization Parameter C

- The parameter C controls the trade-off between maximizing the margin and minimizing classification errors.
- A smaller C value leads to a wider margin but allows more margin violations (left-side figure). Such a model tends to generalize better.
- A larger C value penalizes margin violations more heavily, resulting in fewer violations but a smaller margin (right-side figure). Such a model is more prone to overfitting.
- As $C \rightarrow \infty$, margin violations become extremely costly, and the classifier approaches a hard-margin classifier. In this case, the classifier attempts to satisfy $y(w^T x + b) \geq 1$ for all training samples.
- If the data is not linearly separable, there may be no feasible hard-margin solution. In such cases, soft-margin SVM allows some margin violations, where C controls the penalty for violating the margin constraints.
- Example



Non-Linear SVM Classification

- In real-world applications, most datasets are not linearly separable.
- One approach to handling nonlinear datasets, similar to linear and logistic regression, is to add additional features such as polynomial features. In some cases, this transformation can make the dataset linearly separable in a higher-dimensional space.

SVM Kernels

- To efficiently handle nonlinear datasets, SVMs use a mathematical technique called the kernel trick.
- Linear SVM uses feature values, while a kernel function measures similarity between two data points in an implicitly transformed feature space.
- SVM identifies support vectors during training via learned coefficients α .
- During prediction, the classifier evaluates a weighted sum of kernel-based similarities between the test point and these support vectors to make the final decision.

$$h(x) = \hat{y} = \text{sign}\left(\sum_{i=1}^n \alpha^{(i)} y^{(i)} k(x^{(i)}, x) + b\right)$$

Where x is the new data point,

$x^{(i)}$ is the i^{th} training samples, and $\alpha^{(i)}$ is its corresponding learned coefficient.

Only training points with non-zero $\alpha^{(i)}$ contribute to the decision function. These are called support vectors.

Only points with $\alpha^{(i)} > 0$ matter. These points are called support vectors.

$k(x^{(i)}, x)$ is the kernel function which measures similarity between the two data points $x^{(i)}$ and x in an implicitly transformed feature space.

- In kernel SVM, the training objective remains to maximize the margin and penalize classification errors, but all computations are performed using kernel functions instead of explicit feature vectors.
- As a result, both the hypothesis function and the optimization objective are expressed in terms of kernel functions rather than explicit feature representations.
- This allows SVM to learn nonlinear decision boundaries efficiently.
- The kernel trick achieves the effect of mapping data into a higher-dimensional feature space without explicitly computing the transformation, avoiding computational explosion.
- Different kernels use different similarity functions.
- Not all similarity functions are valid kernels. Valid kernels must satisfy certain mathematical conditions (such as Mercer's condition) to ensure that the optimization problem remains convex and solvable.

Types of Kernels

linear kernel used for already linear data

Linear Kernel

- A linear kernel does not transform the feature space and produces a standard linear SVM classifier.
- It is suitable when the dataset is approximately linearly separable.

Polynomial Kernel

- The polynomial kernel creates nonlinear decision boundaries by introducing polynomial relationships between features.
- The polynomial kernel function is:

$$k(x, l) = (x^T l + c)^d$$

Where x and l are the two data points, d is the degree of the polynomial, and c is a constant hyperparameter.

- Higher polynomial degrees allow more complex decision boundaries but may increase the risk of overfitting.

Gaussian Radial Basis Function (RBF) Kernel

- The Gaussian Radial Basis Function (RBF) kernel is one of the most commonly used kernels in SVMs.
- It measures similarity between data points based on their distance from each other.
- The RBF kernel function is:

$$k(x, l) = \exp\left(-\frac{\|x - l\|^2}{2\sigma^2}\right) \text{ or } k(x, l) = \exp(-\gamma\|x - l\|^2)$$

Where σ^2 is the variance and a hyperparameter; $\gamma = 1/2\sigma^2$.

- If two points are close together, the similarity value becomes close to 1. If they are far apart, the similarity approaches 0.
- A small value of γ (large σ^2) produces smoother decision boundaries and better generalization.
- A large value of γ (small σ^2) produces more complex decision boundaries and may lead to overfitting.

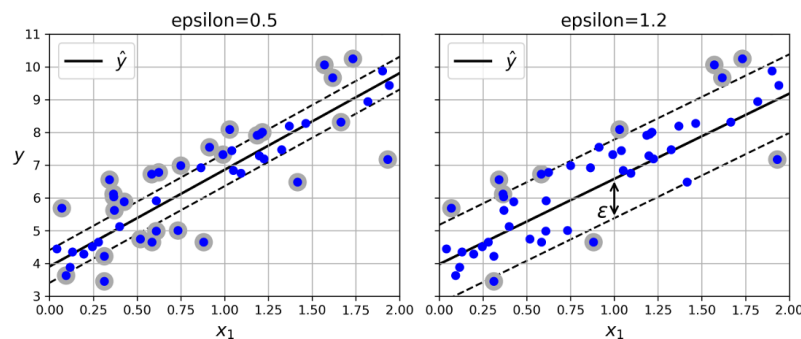
Other Kernels

- Sigmoid kernel, String kernel (used in text classification), Chi-square kernel

Support Vector Regression (SVR)

- Support Vector Machines can also be used for regression tasks. The regression version of SVM is called Support Vector Regression (SVR).
- SVR introduces a margin of tolerance around the regression function, called the epsilon-insensitive tube.
- Instead of maximizing the margin between two classes, SVR tries to fit as many training samples as possible within this margin of tolerance around the regression function, while limiting prediction errors outside this margin.
- The width of the margin/tube is controlled by a hyperparameter, ϵ .
- **Example**

The following figure shows two linear SVM regression models trained on some linear data, one with a small margin ($\epsilon = 0.5$) and the other with a larger margin ($\epsilon = 1.2$).



Loss Function in SVR

- The objective of SVR is not to minimize every prediction error. Instead, it tries to ensure that prediction errors remain within a specified tolerance level.
- If the predicted value lies within this tube, the error is ignored and no penalty is applied.
- Only points lying outside the tube contribute to the loss function. These points are called support vectors.
- The width of the tube is controlled by the hyperparameter ϵ .
 - A larger value of ϵ creates a wider tube and allows more prediction error, resulting in a smoother model.
 - A smaller value of ϵ creates a narrower tube and attempts to fit the data more closely.

- SVR uses the epsilon-insensitive loss function. For i^{th} example it is given as:

$$L(y^{(i)}, \hat{y}^{(i)}) = \max(0, |y^{(i)} - \hat{y}^{(i)}| - \epsilon)$$

- If the prediction error is smaller than ϵ , the loss becomes zero.
- If the prediction error exceeds ϵ , a positive loss is incurred.

- Hence the overall objective function is

$$J(w, b) = \text{minimize } \frac{1}{2} \|w\|^2 + C \sum_{i=1}^n \max(0, |y^{(i)} - \hat{y}^{(i)}| - \epsilon)$$

- Similar to classification SVM, the parameter C controls the trade-off between model complexity and prediction error.
 - A small value of C allows more errors and produces a smoother regression function.
 - A large value of C penalizes errors more strongly and attempts to fit the training data more closely.

- SVR can also use kernel functions.
- A linear SVR produces a linear regression function, whereas nonlinear kernels such as the Gaussian RBF kernel allow SVR to model complex nonlinear relationships.